

BGP Security: How Does BGP Work and the Threat of Prefix Hijacks

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SprintML Group at CISPA

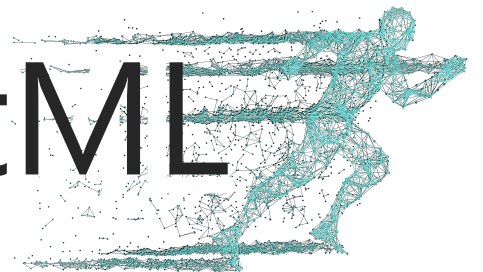
April 24th, 2025



CISPA

HELMHOLTZ CENTER FOR
INFORMATION SECURITY

SprintML



Lecture Plan on BGP Security



1. **BGP (Border Gateway Protocol)** is the routing protocol of the Internet.



2. **Prefix Hijack** attacks maliciously re-route the Internet traffic.



3. **Destructive attack:** real-world incidents lead to internet outage, disrupted services, lost money, etc.



4. **Securing BGP:** current efforts and challenges with a focus on RPKI (Resource Private Key Infrastructure) and BGPsec.

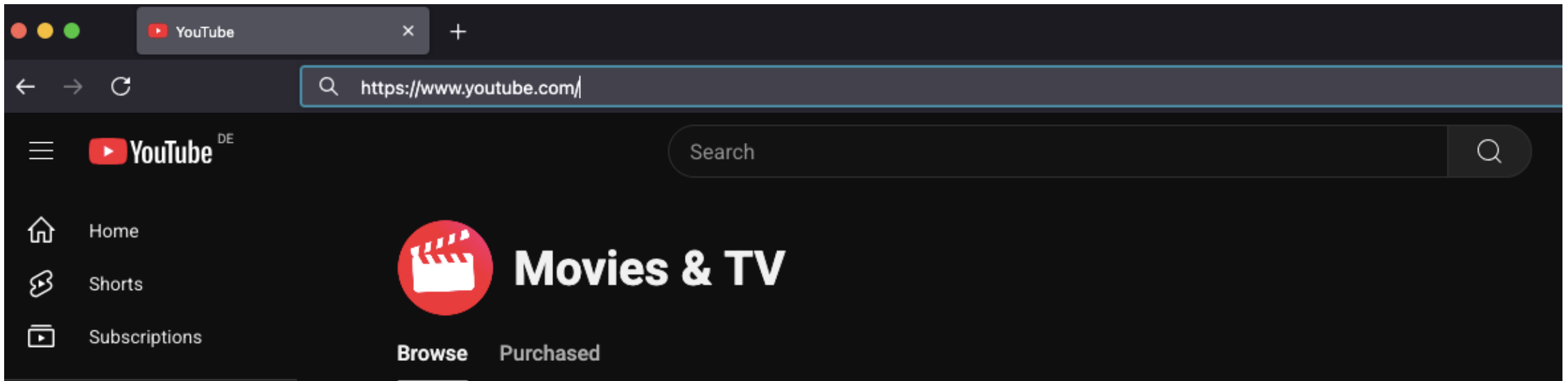
How do we connect to YouTube?



TU Darmstadt



user



The global Internet as a Network of Networks

AS (Autonomous System) a large network run by an organization.



130.83.0.0/16 IP prefix

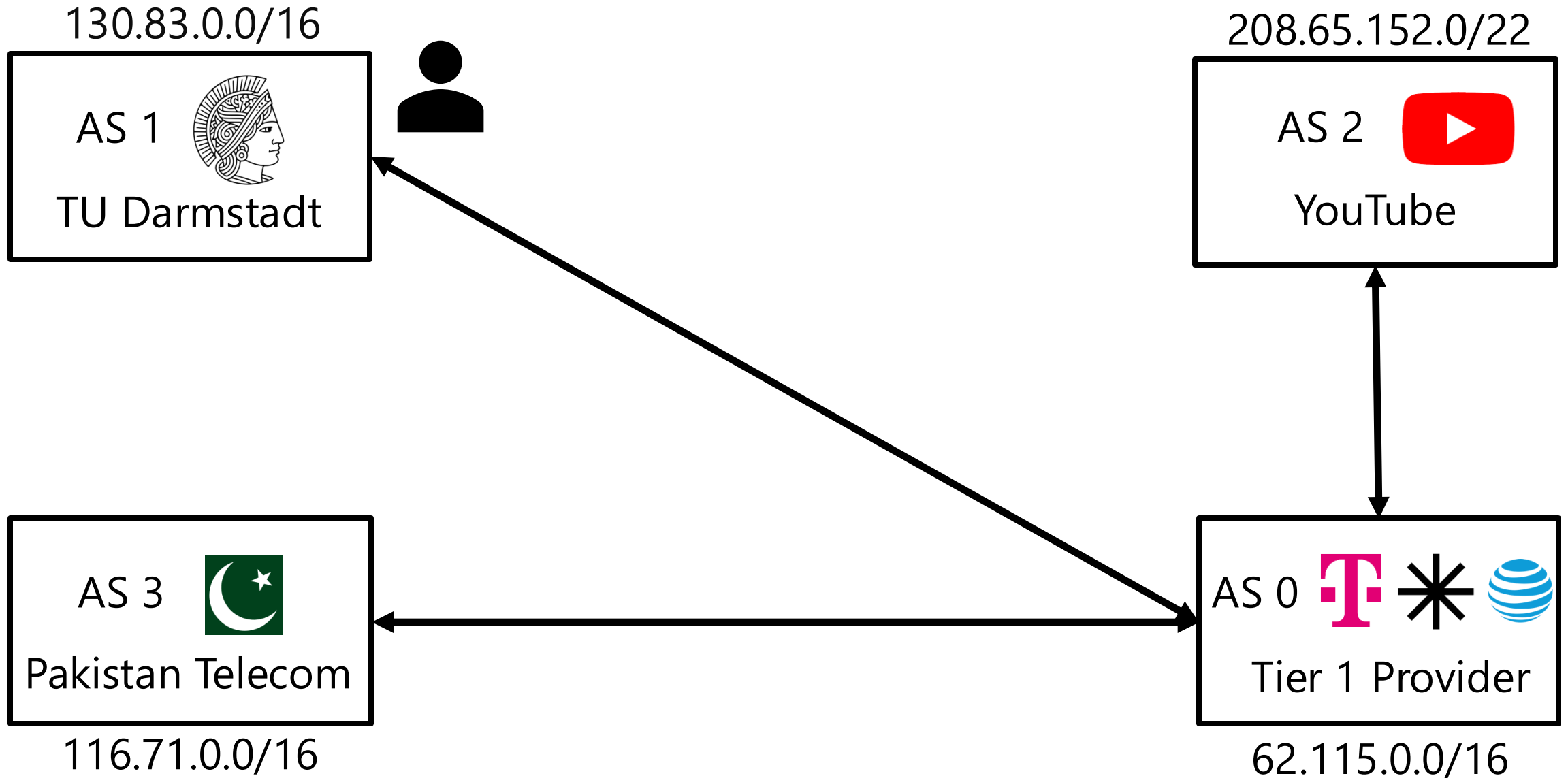
network
address

/16 prefix
length

IP Ranges		AS8365		IPv4 Ranges	IPv6 Ranges
NETBLOCK	COMPANY		NUM OF IPS		
<u>130.83.0.0/16</u>	 TU Darmstadt, Hochschulrechenzentrum		65,536		

The global Internet as a Network of Networks

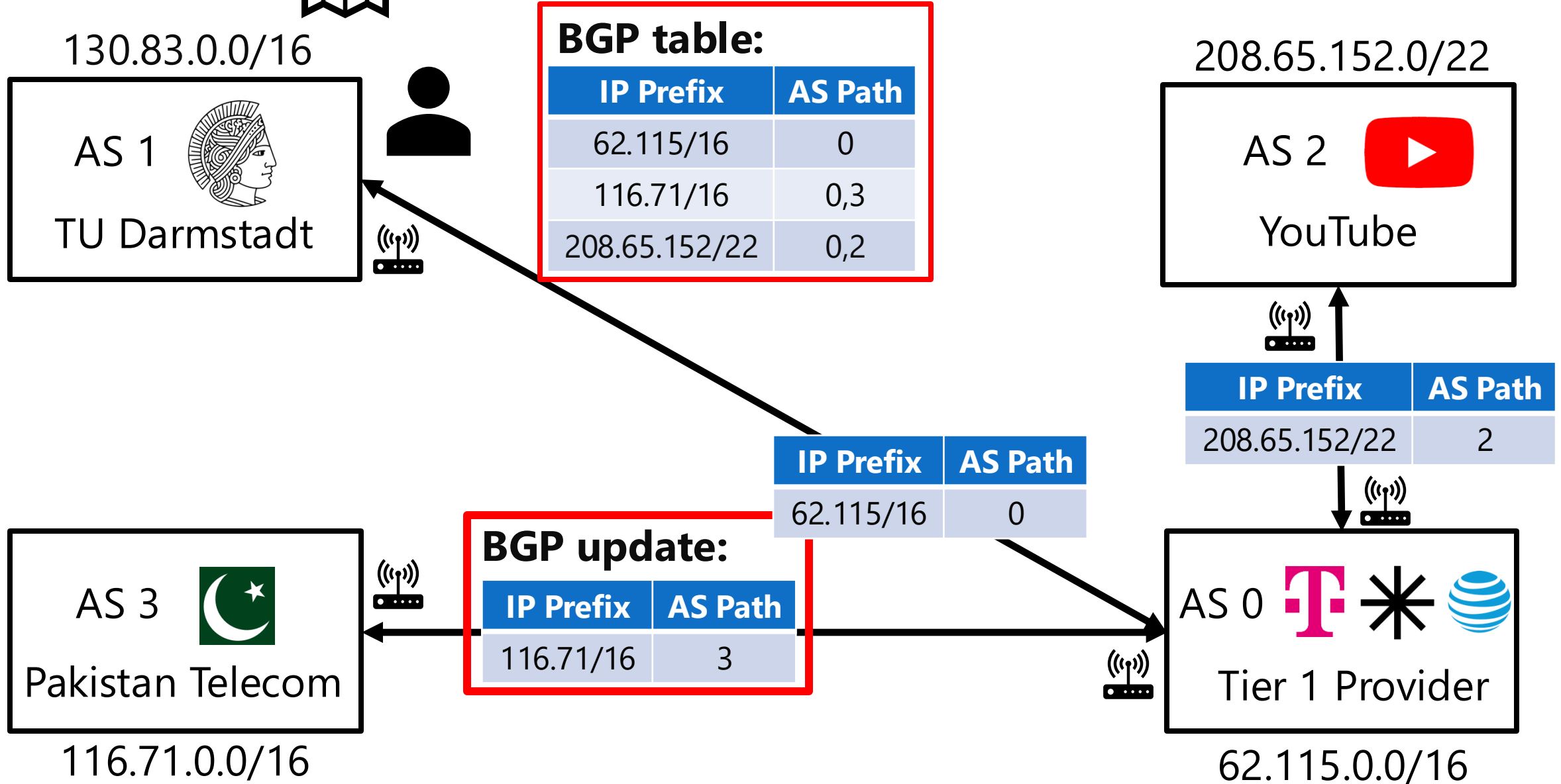
The Internet is a mesh of connected **ASes** (Autonomous Systems)



BGP (Border Gateway Protocol)



BGP routes the Internet traffic.



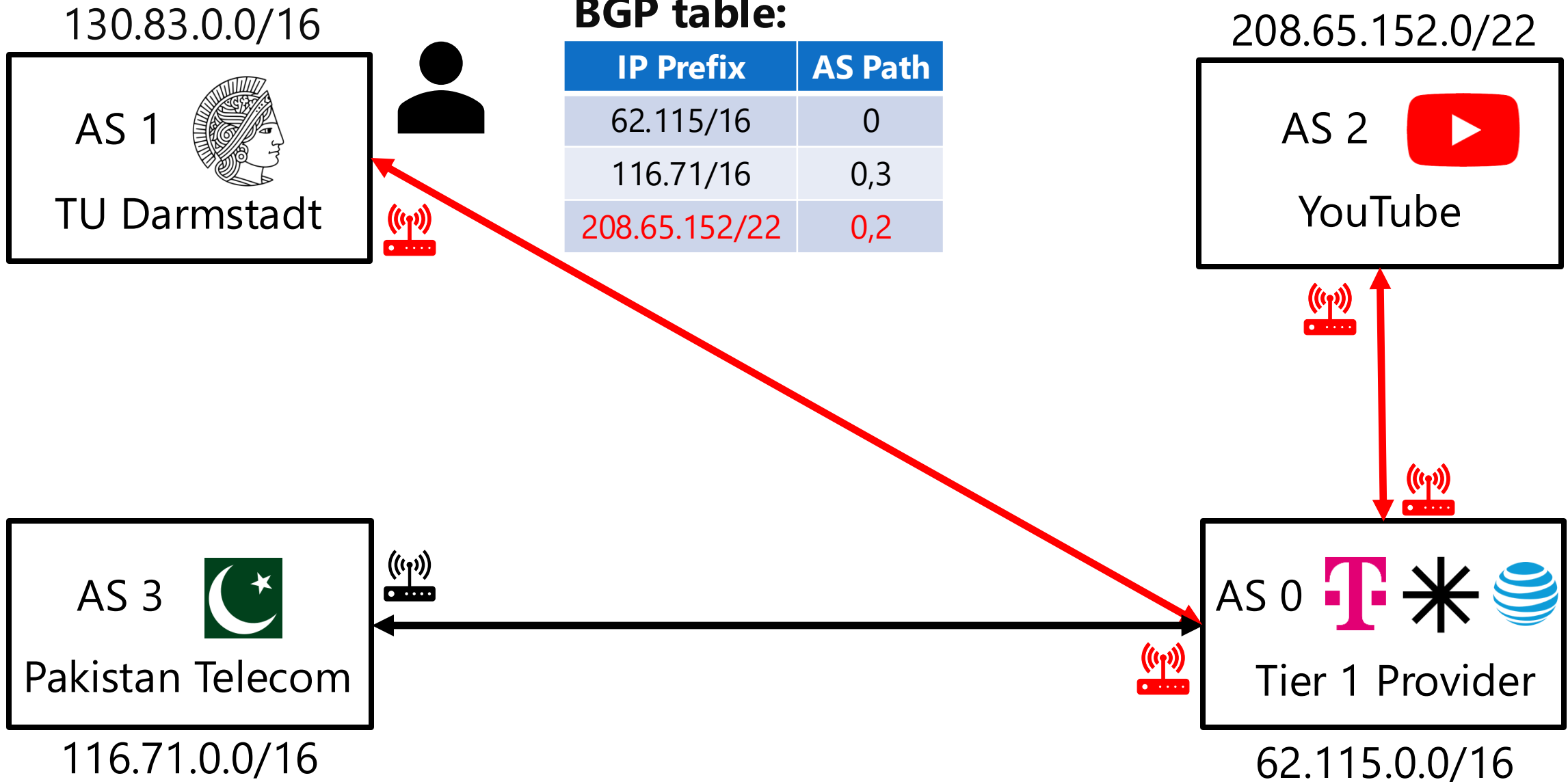
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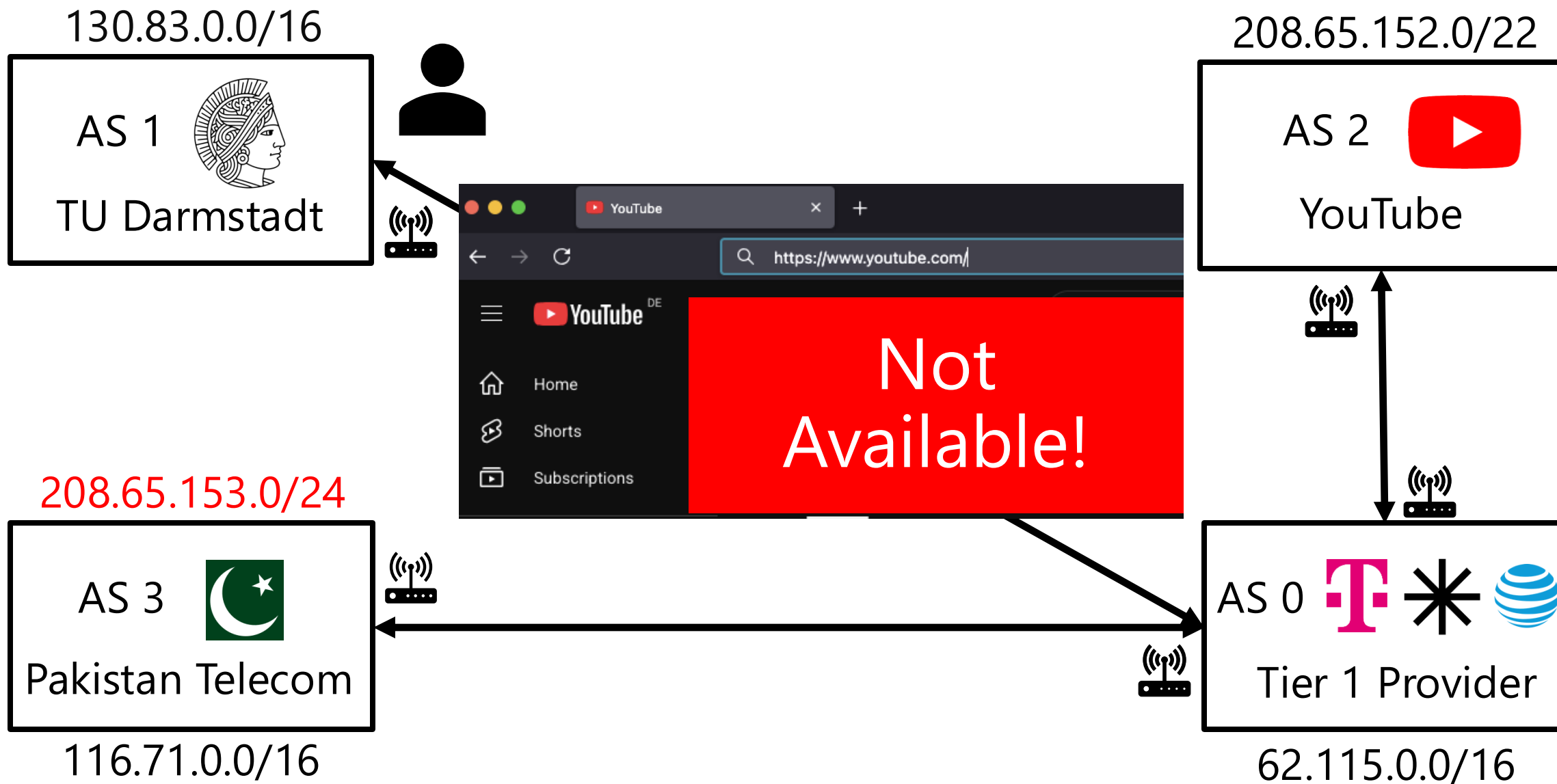
BGP table:

IP Prefix	AS Path
62.115/16	0
116.71/16	0,3
208.65.152/22	0,2



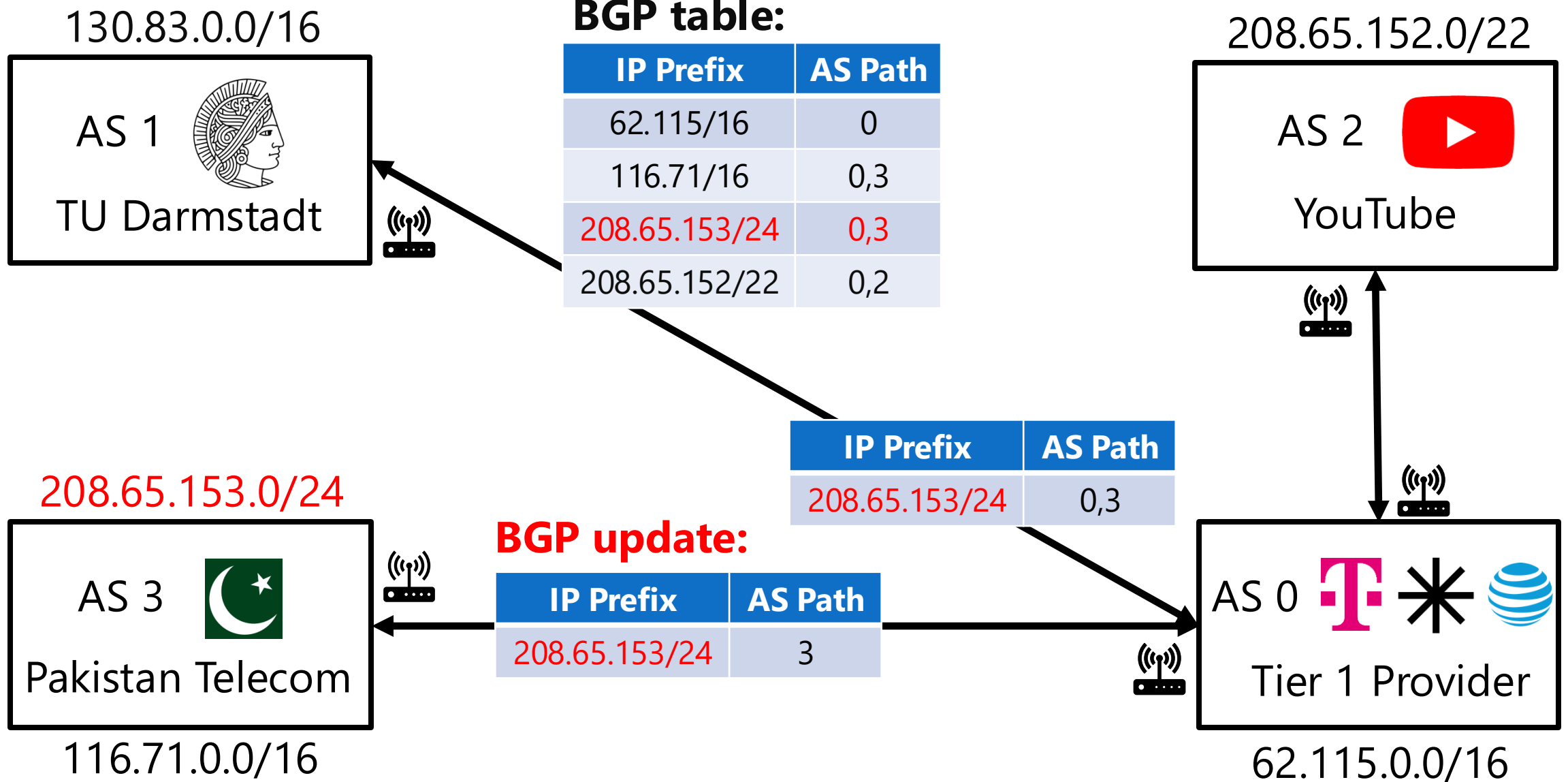


YouTube no longer available!

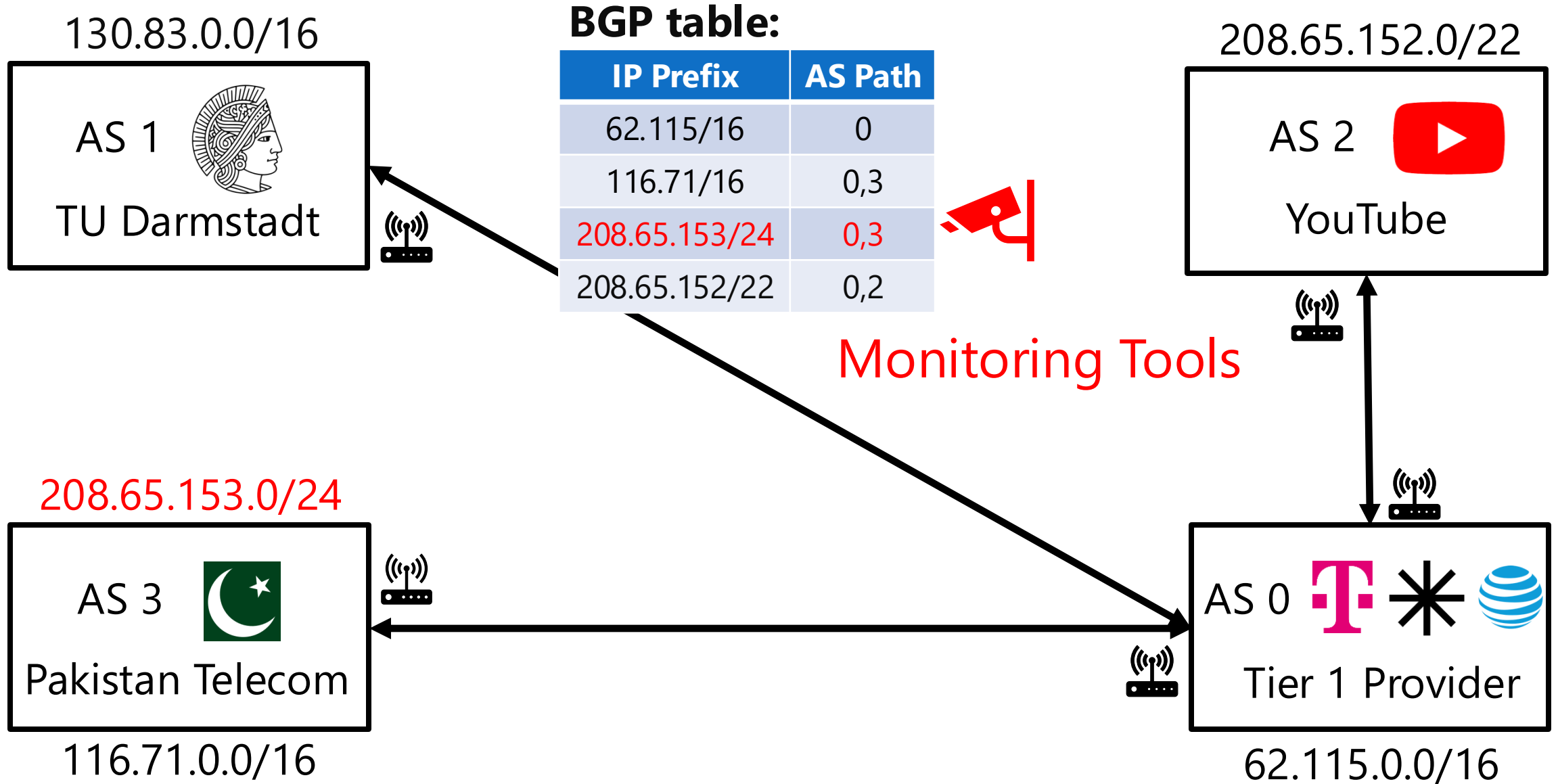


Prefix Hijack: AS falsely announces a prefix

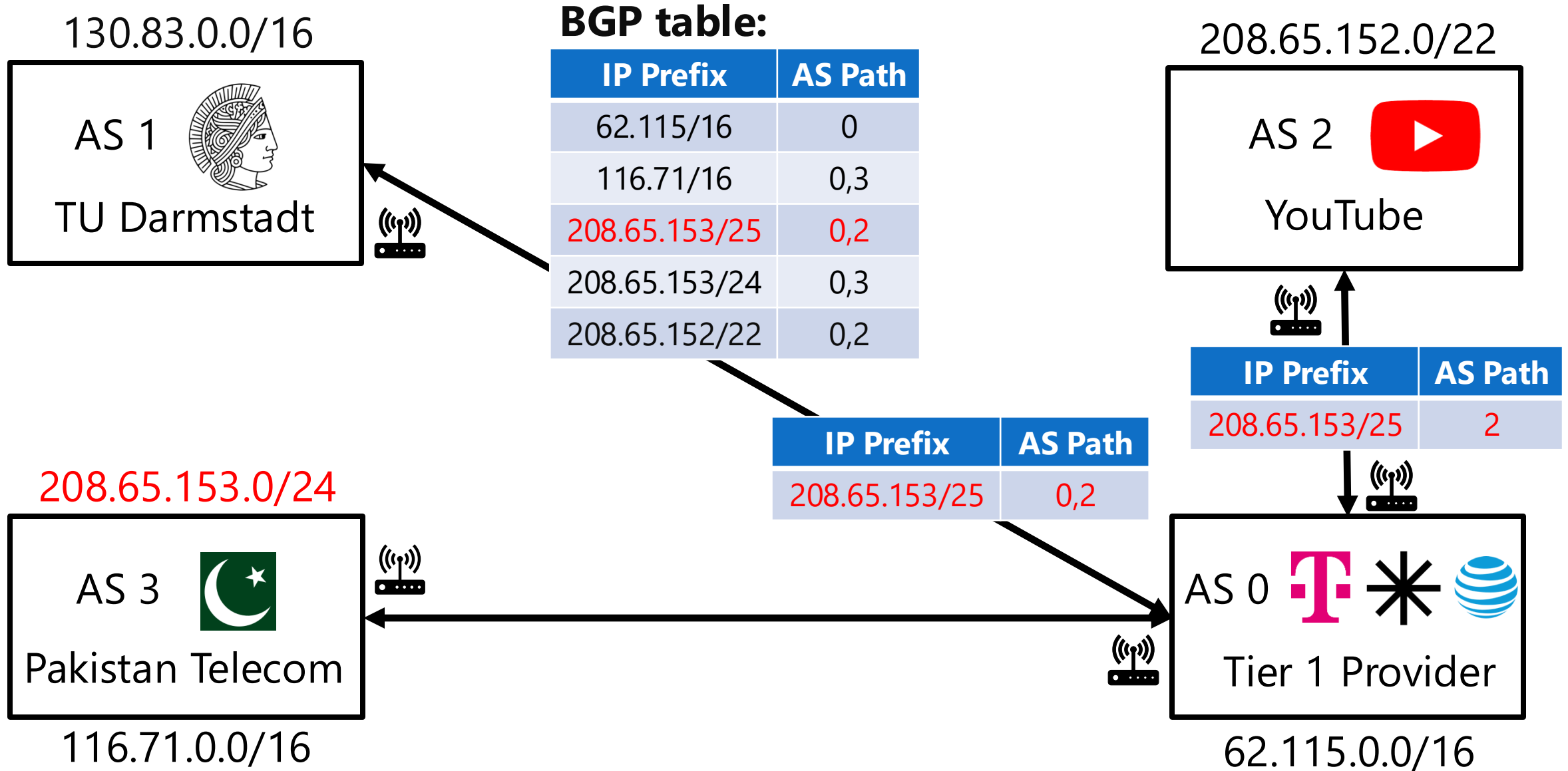
🦋 Pakistan Telekom announces IP (sub)prefix that belongs to YouTube!



Prefix Hijack detected with Monitoring Tools

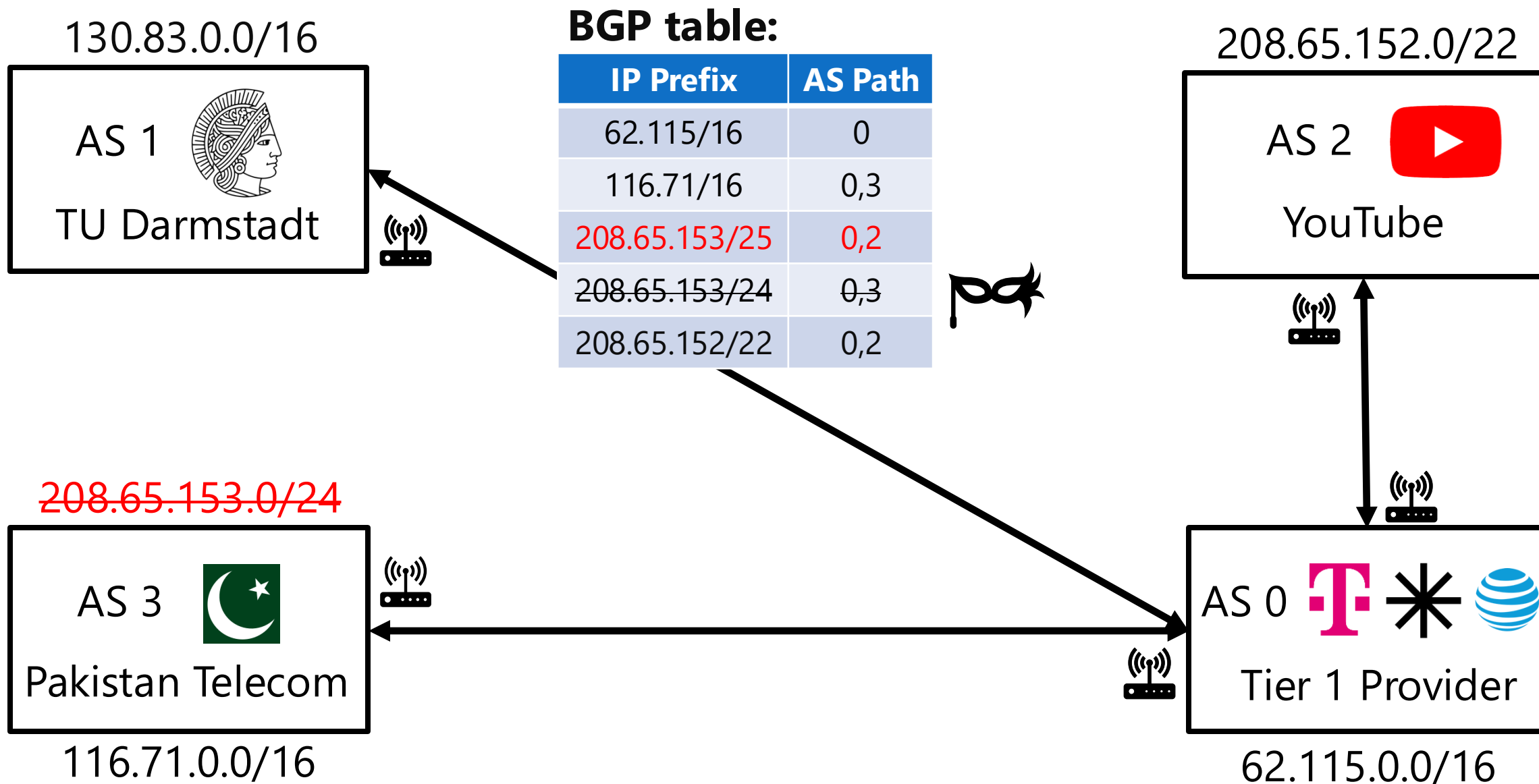


YouTube patch with more specific (sub)prefix





Announce invalid route for prefix filtering



Prefix Hijack interrupted YouTube for hours



YouTube Hijacking: A RIPE NCC RIS case study

17 Mar 2008 · internet governance news RIS

**YouTube
interrupted
for hours!**



Introduction

On Sunday, 24 February 2008, Pakistan Telecom (AS17557) started an unauthorised announcement of the prefix 208.65.153.0/24. One of Pakistan Telecom's upstream providers, PCCW Global (AS3491) forwarded this announcement to the rest of the Internet, which resulted in the hijacking of YouTube traffic on a global scale.

In this report we show how the events were seen by RIPE NCC's [Routing Information Service \(RIS\)](#) and how, in general, one can use the RIS tools to obtain hard data on network events.

\$1.9 million stolen in a Prefix Hijack in 2022!

YouTube Hijacking: A RIPE NCC RIS case study

17 Mar 2008 · internet governance news RIS

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News

Cybercrime

Technology

YouTube interrupted for hours!



\$1.9 million stolen in a Prefix Hijack!

February 14th, 2022

KlaySwap crypto users lose funds after BGP hijack

Hackers have stolen roughly \$1.9 million from South Korean cryptocurrency platform **KLAYswap** after they pulled off a rare and clever BGP hijack against the server infrastructure of one of the platform's providers.

More than 84000 Prefix Hijacks in one week!

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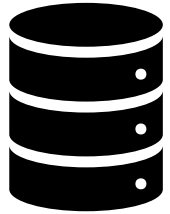
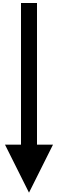
RPKI: Resource Public Key Infrastructure



RPKI Repository

(e.g., *Internet Registry)

stores



ROAs (Route Origin
Authorizations)

208.65.152.0/22

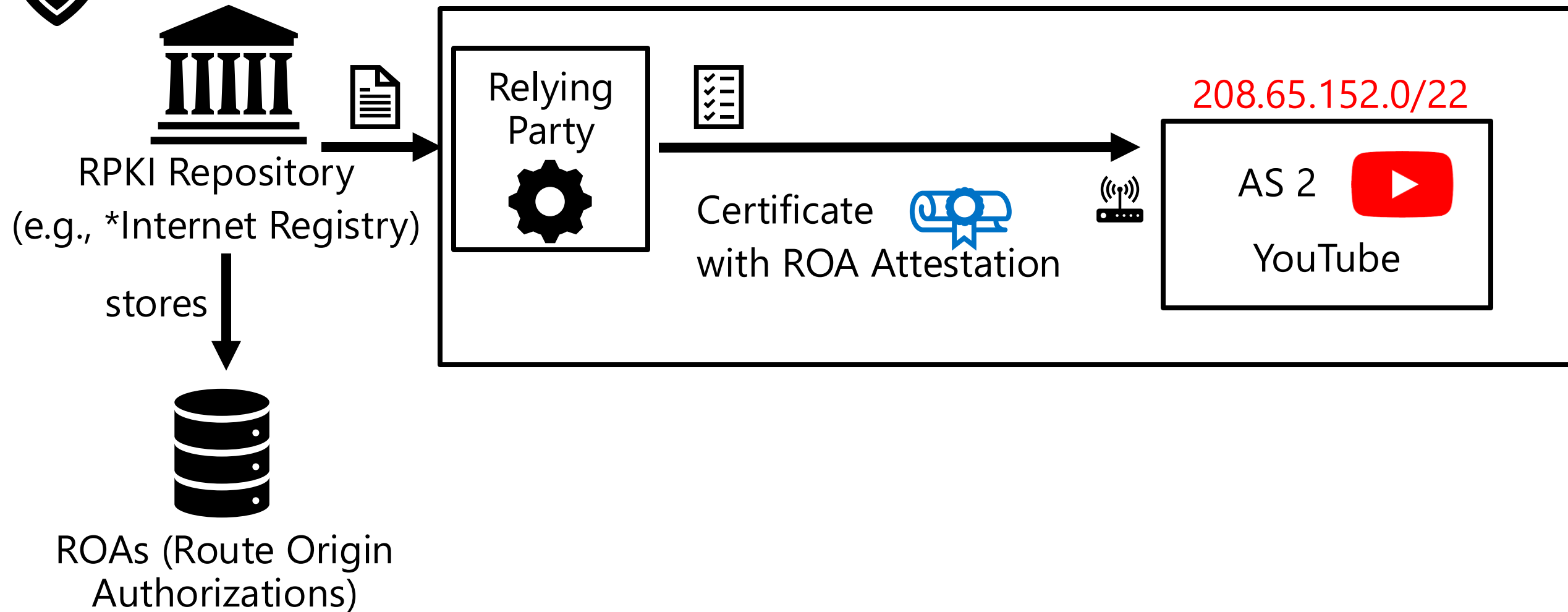
AS 2



YouTube

IP Prefix	Authorized AS	Sign
208.65.152.0/22	2	
...	...	

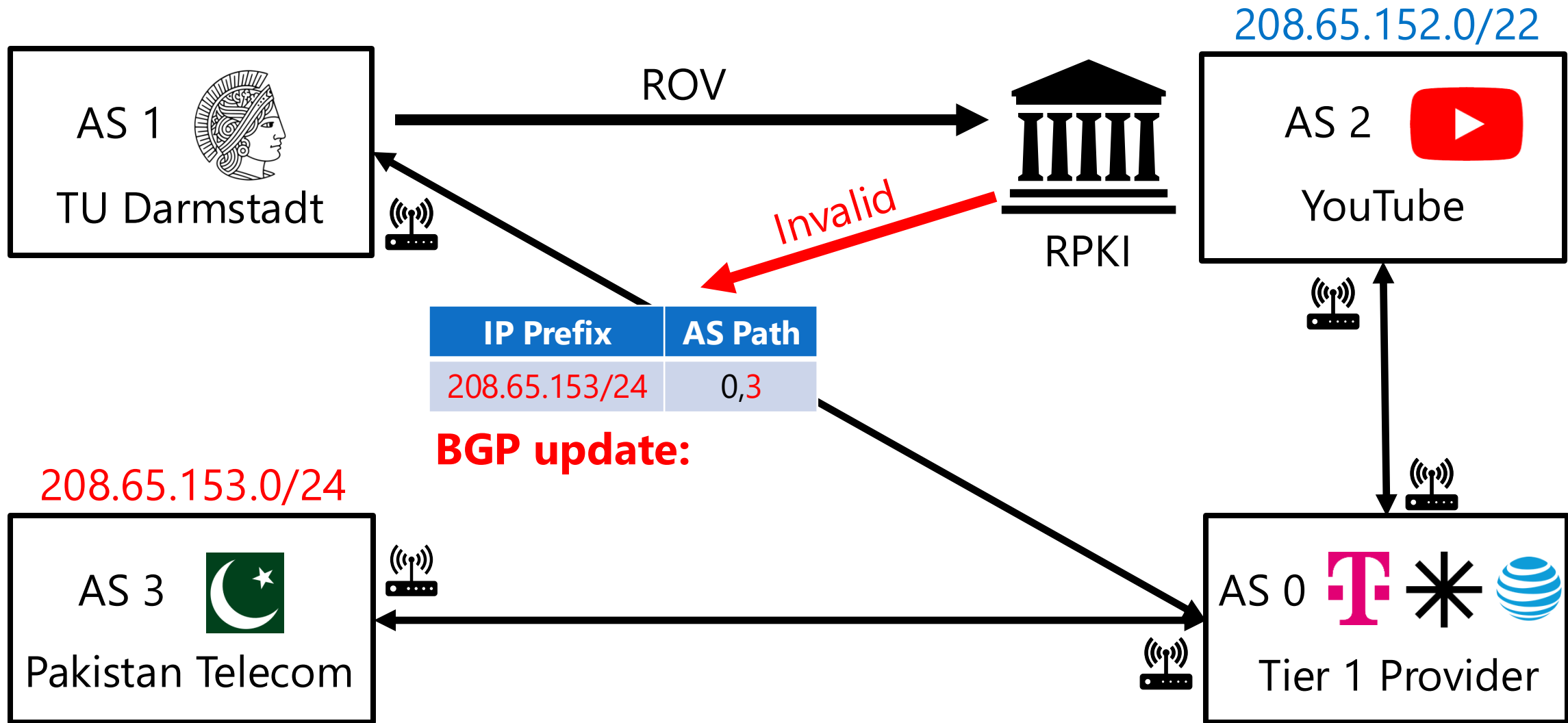
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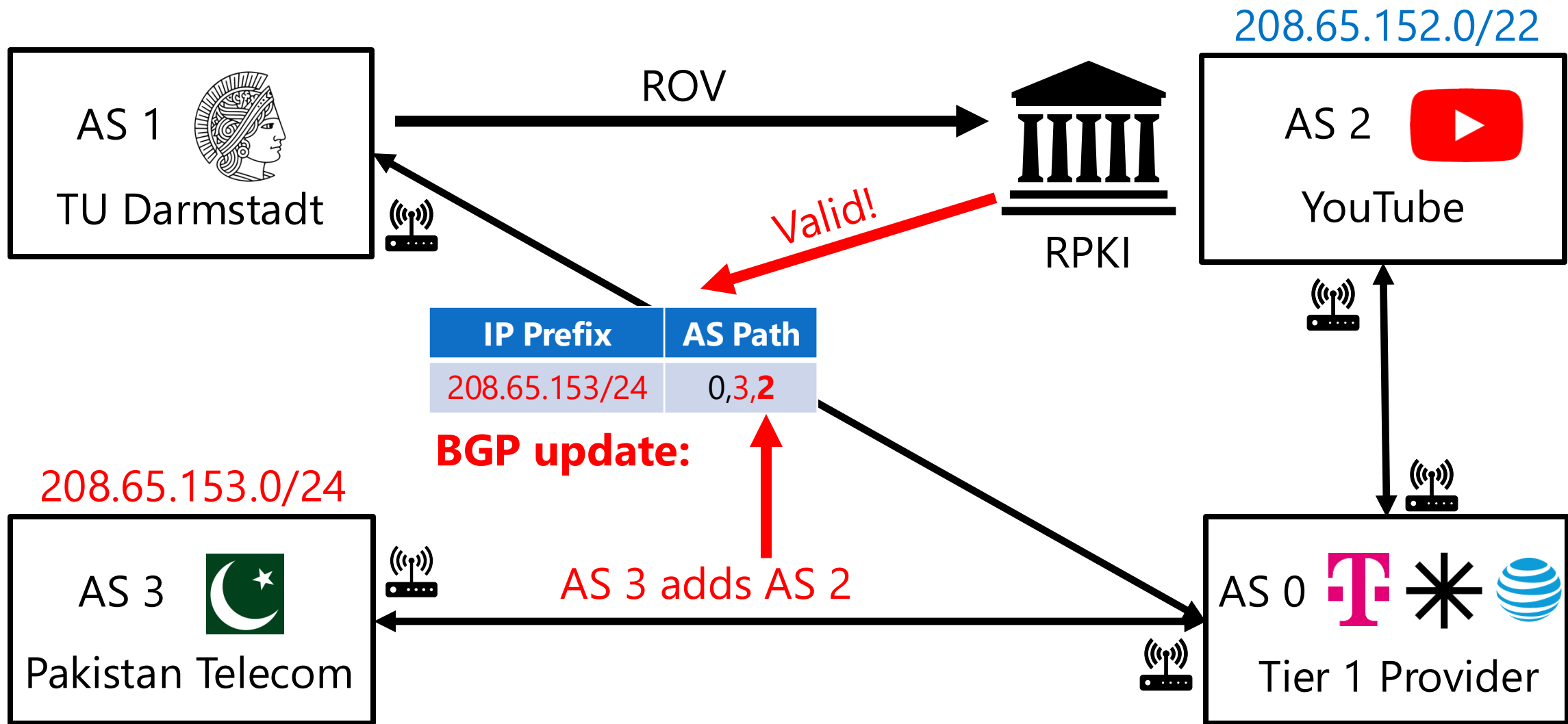
RPKI with ROV (Route Origin Validation)

ROV rejects IP prefixes originated by an **unauthorized** AS.



ROV does not verify the AS path in BGP

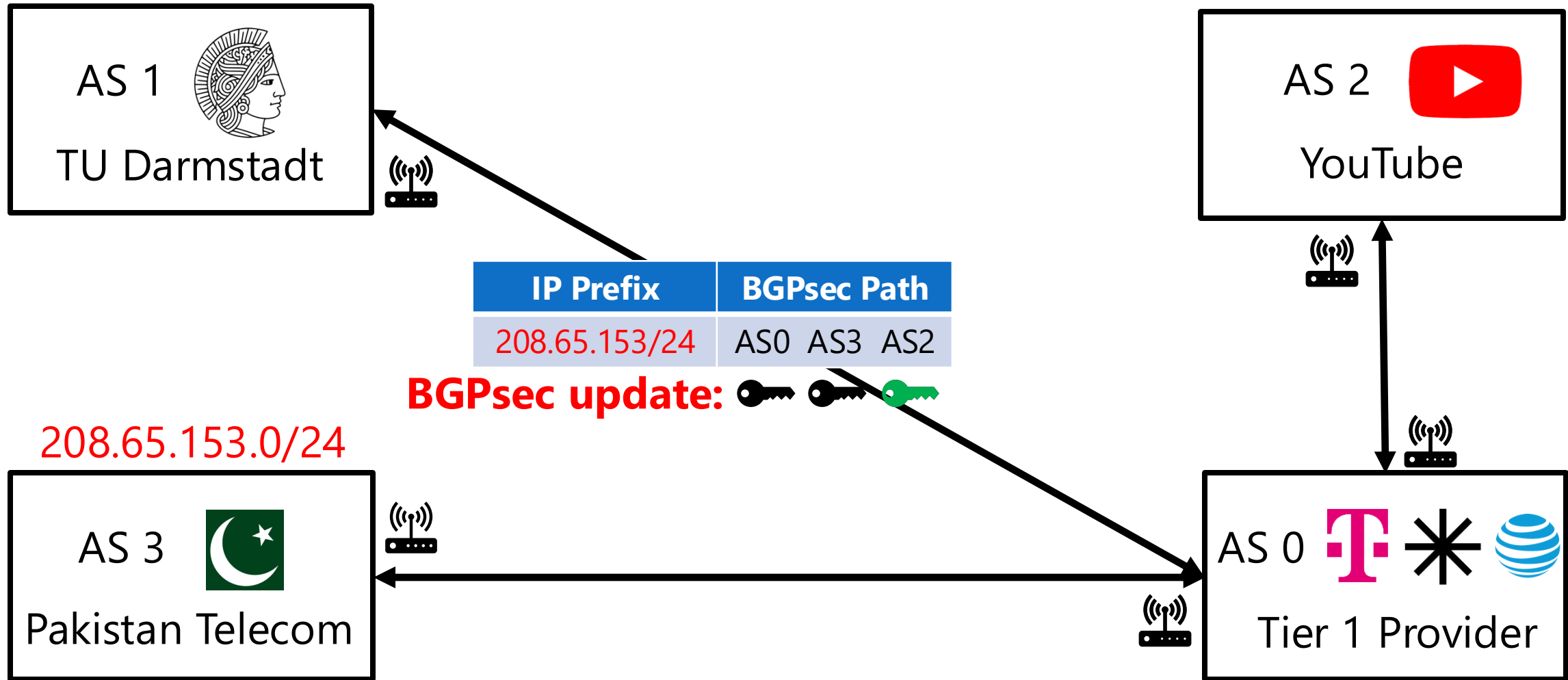
ROV accepts IP prefixes originated by an **authorized** AS.





BGPsec: verification of the AS Path

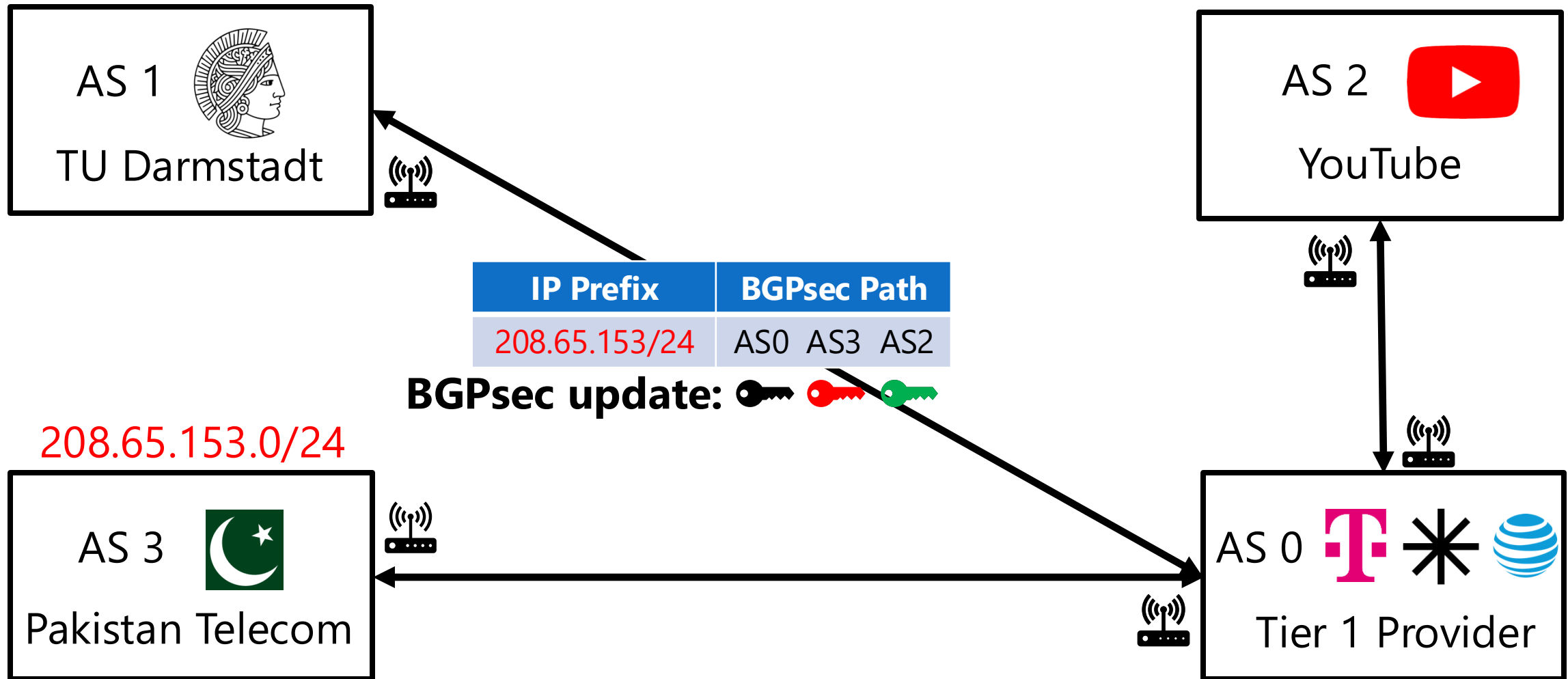
Valid signature: AS 2 owns the prefix!



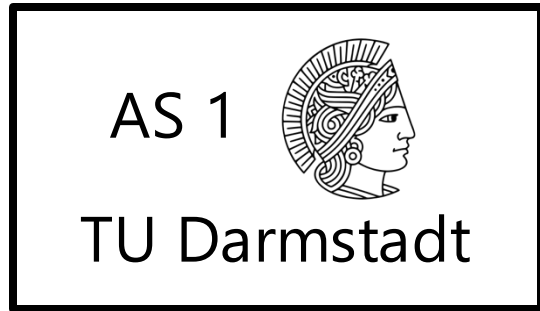


BGPsec: verification of the AS Path

Invalid BGPsec Path: AS2 did not send the BGPsec update to AS3!



Deployment of BGPsec is resource-intensive



IP Prefix	BGPsec Path
208.65.153/24	AS0 AS3 AS2

BGPsec update:   



High-cost of BGPsec updates:

1. Large size of the updates due to signatures.
2. Costly check of signatures for each hop in the BGPsec Path.
3. Special hardware required to speed up the signature check.
4. Many more BGPsec than BGP updates.

Approaches to Secure BGP: Pros & Cons

Approach	Security Action	Pros	Cons
Monitoring Tools	Anomaly detection	Detects hijacks	Reactive
Prefix Filtering	Announce invalid routes	Simple implementation	Manual maintenance
Resource PKI	Origin AS authenticity	More adaptation	Insecure AS path
BGPsec	Secure AS path	Very secure	High overhead

Main challenge: balance between network performance & security.

Conclusions on BGP and its Security

-  1. **BGP** (Border Gateway Protocol) is a traffic routing protocol of the Internet that connects ASes (Autonomous Systems).
-  2. **Prefix Hijack**: AS falsely announces an IP prefix that it does not own or an incorrect AS path: detect by monitoring tools.
-  3. **Destructive attack**: prefix hijack leads to internet outage, disrupted services, lost money, etc.
-  4. **Securing BGP**: RPKI checks origin AS and BGPsec verifies the AS path; 12.3% of ASes use ROV (Route Origin Verification)!

Additional Materials

YouTube Video

Lecture Slides

1. BGP: Chapter 4 in the book "Computer Networks: A Systems Approach": <https://book.systemsapproach.org>
2. YouTube incident: <https://www.ripe.net/about-us/news/youtube-hijacking-a-ripe-ncc-ris-case-study/>
3. Lost cryptocurrency: <https://blog.cloudflare.com/bgp-leaks-and-crypto-currencies/>
4. RPKI: <https://www.noction.com/blog/rpki-overview>
5. BGPsec: <https://www.noction.com/blog/bgpsec-protocol>

Prompts for the Discussion

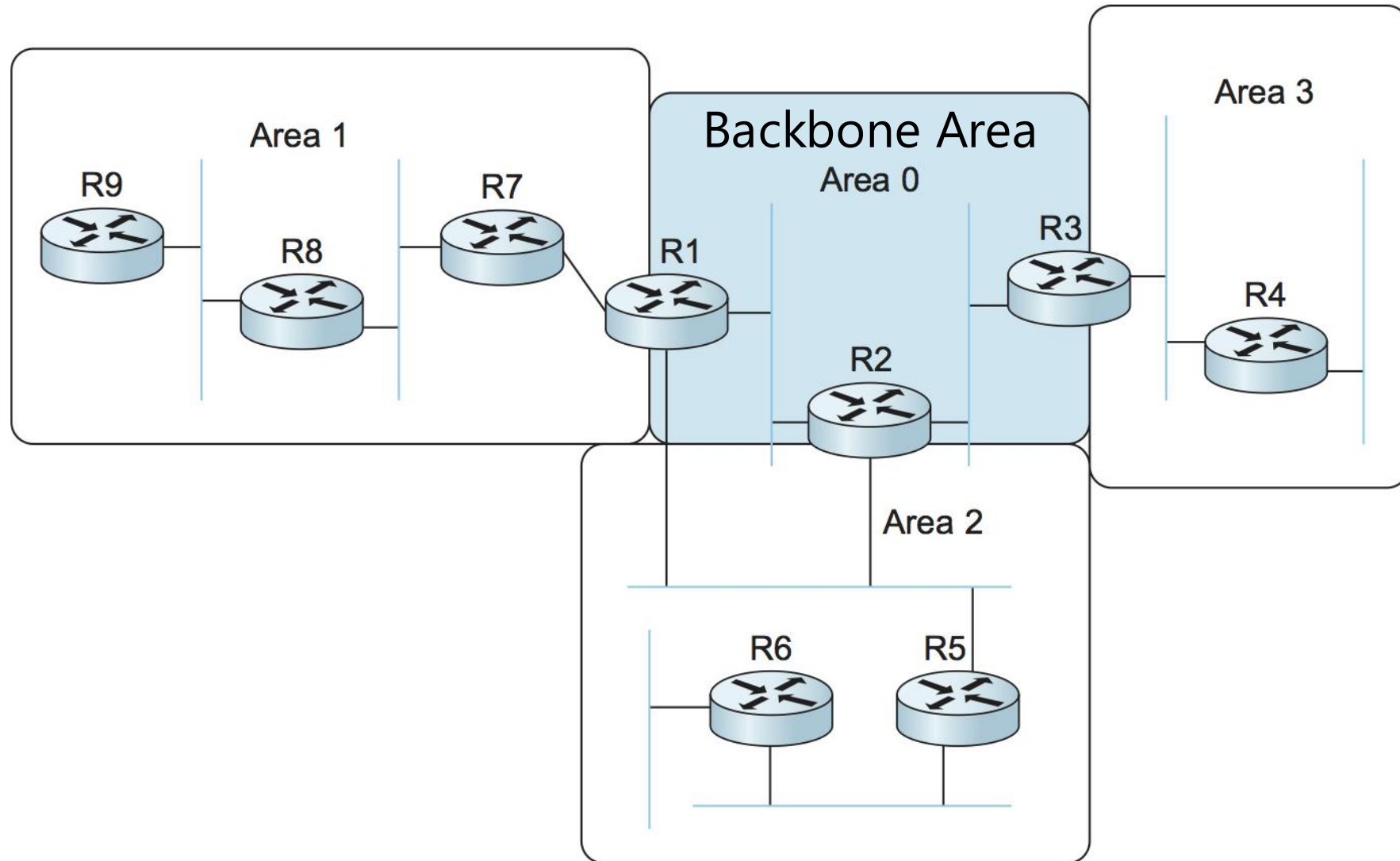
1. What might help small ISPs (Internet Service Providers) adopt RPKI (Resource Public Key Infrastructure) or BGPsec?
2. How much overhead can we tolerate to deploy RPKI or BGPsec?
3. Should governments or EU (European Union) enforce BGP security?

BGP: Priority in routing (path selection)

1. Prefer more specific IP prefix (longest prefix match); Higher Local Preference (LOCAL_PREF, weight).
2. Is the path locally originated?
3. Shortest AS Path (fewer AS hops is preferred).
4. Compare the Multi-Exit Discriminator (MED): the path with the lowest MED is preferred
5. The path with the lowest MED is preferred.
6. Lowest IGP metric to next-hop (lower internal cost to reach the BGP next-hop).
7. Oldest route - among equal options, the oldest (most stable) is preferred.
8. Lowest Router ID (used as a final tiebreaker).

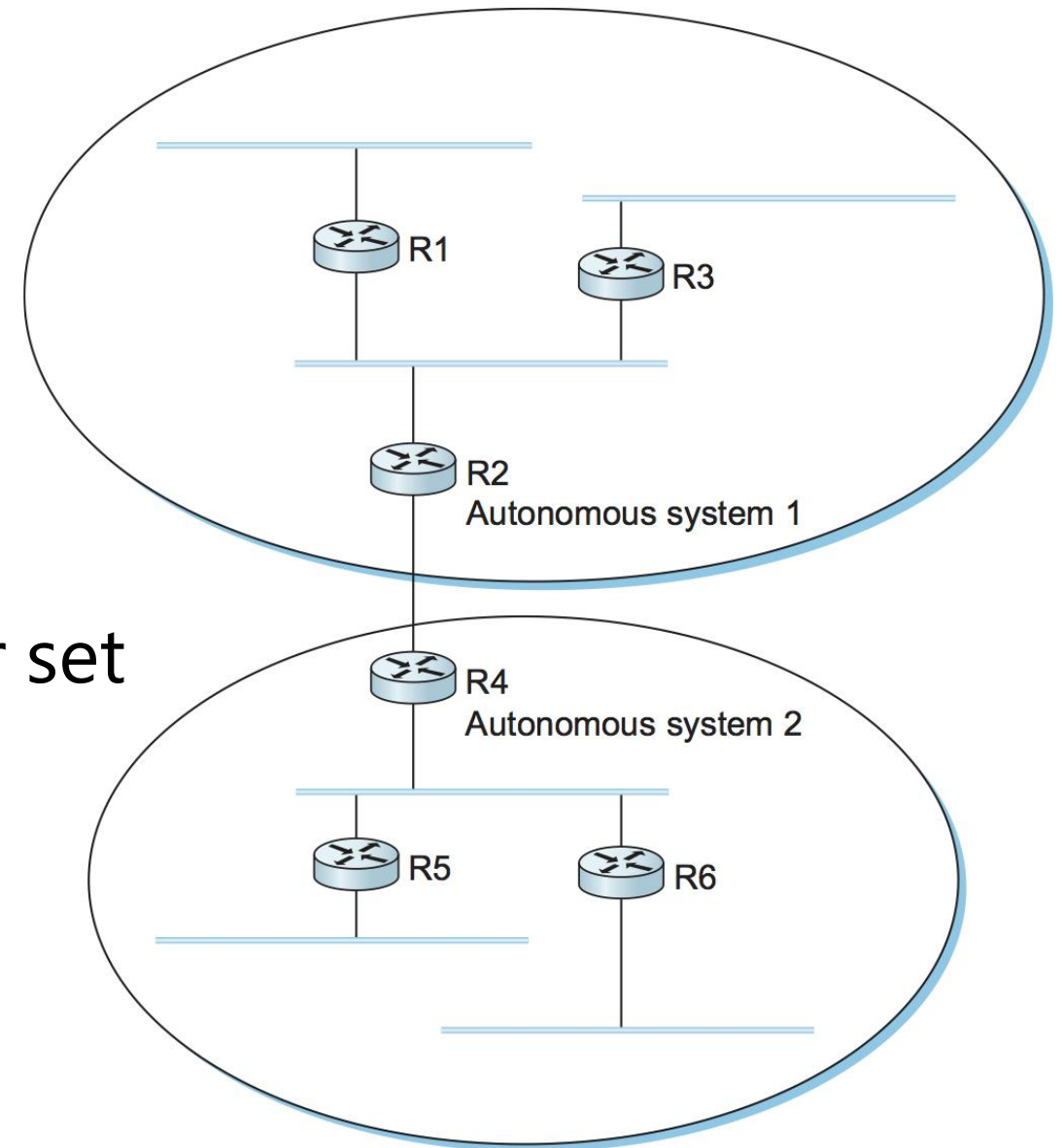
Source: <https://community.cisco.com/t5/networking-knowledge-base/understanding-bgp-best-path-selection-manipulation/ta-p/3150576>

Routing Areas withing a Routing Domain



Autonomous System == Routing Domain

Interdomain Routing: Autonomous Systems (Routing Domains) share their set of IP addresses.



Tier 1 Providers

Name ⇅	Headquarters ⇅	AS number ⇅	CAIDA AS rank ^[16] ▲	Fiber route (km) ⇅	Peering policy ⇅
Lumen Technologies (formerly CenturyLink , formerly Level 3) ^{[27][28][29]}	 United States	3356	1	885,139 ^{[30][31]}	Lumen Peering Policy ↗
Arelion (formerly Telia Carrier) ^[38]	 Sweden	1299	2	70,000 ^[39]	Arelion's IP Network Peering Policy 
GTT Communications	 United States	3257	4	232,934 ^{[21][22]}	GTT Peering Policy ↗
NTT Communications (formerly Verio) ^[32]	 Japan	2914	5	?	Global Peering Policy ↗
Tata Communications (prev. VSNL prev. Teleglobe) ^[35]	 India	6453	6	700,000 ^[36]	Peering Policy ↗
Telecom Italia Sparkle (Seabone) ^[37]	 Italy	6762	8	560,000	Peering Policy ↗
Zayo Group (formerly AboveNet) ^[48]	 United States	6461	9	196,339 ^[49]	Zayo Peering Policy ↗
PCCW Global	 Hong Kong	3491	10	?	Peering policy ↗
Orange ^[33]	 France	5511	11	495,000 ^[34]	OTI peering policy ↗
Telxius (Subsidiary of Telefónica) ^[40]	 Spain	12956	15	65,000 ^[41]	Peering Policy 
Deutsche Telekom Global Carrier ^[19]	 Germany	3320	17	250,000 ^[20]	DTAG Peering Details ↗
Verizon Enterprise Solutions (formerly UUNET) ^[46]	 United States	701	25	805,000 ^[47]	Verizon UUNET Peering policy 701, 702, 703 ↗
Liberty Global ^{[23][24]}	 Netherlands ^[25]	6830	26	800,000 ^[26]	Peering Principles ↗
AT&T ^[17]	 United States	7018	29	660,000 ^[18]	AT&T Peering policy ↗

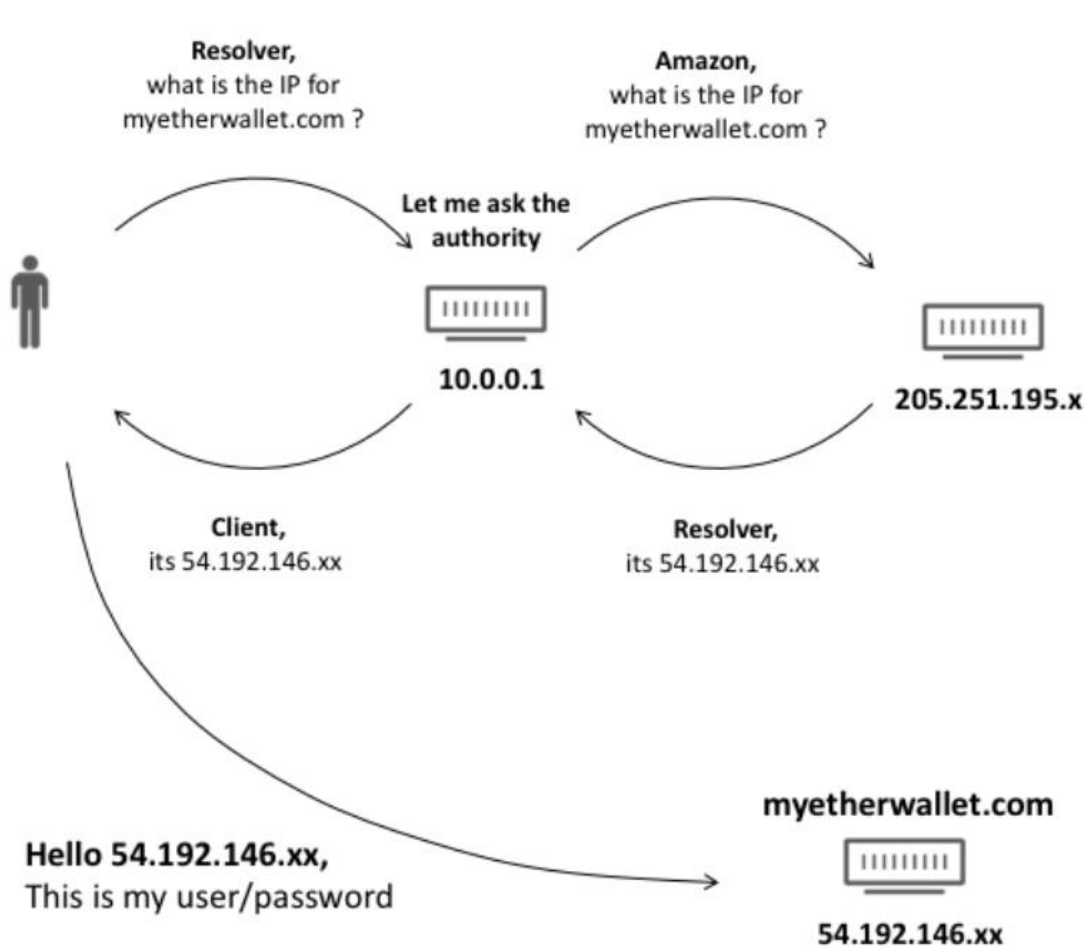
Background Information: DNS, BGP, etc.

DNS (domain name system) servers provide the IP address, but BGP provides the most efficient way to reach that IP address. Roughly speaking, if DNS is the Internet's address book, then BGP is the Internet's road map.

Because Internet is decentralized, there is no governing body or traffic cop laying down the best routes for data packets to travel to their intended IP address destinations. BGP fulfills this role.

Cases for BGP: Normal vs Prefix Hijacking

Normal Case



Prefix Hijacking

